Lecture-2
Prof. Dr. Nurul Huda
Computational Linguistics

Application of DFA

- Web search
- Extraction of information from text

Finding Strings in Text

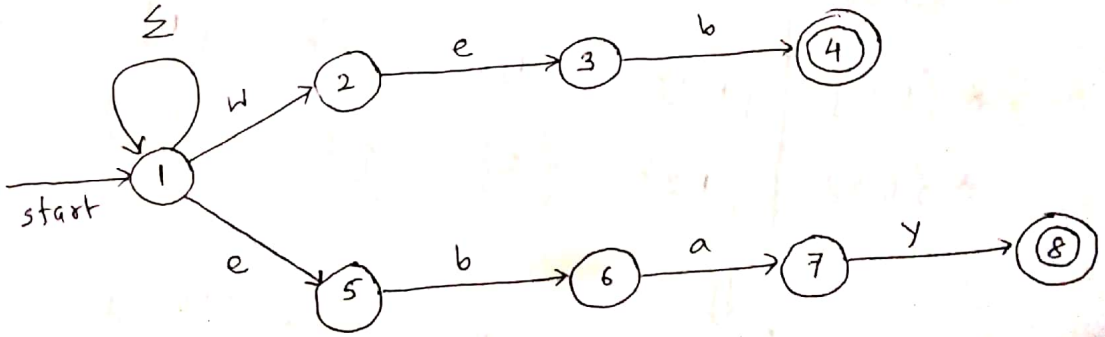
Given a set of words, find all documents that contain one (or all) of those words.

Example: A search engine is a popular examples of this process.

Non-deterministic Finite Automata for Text search:

Ques: Suppose we want to design a NFA to recognize occurrences of the words web and ebay.

Ans:



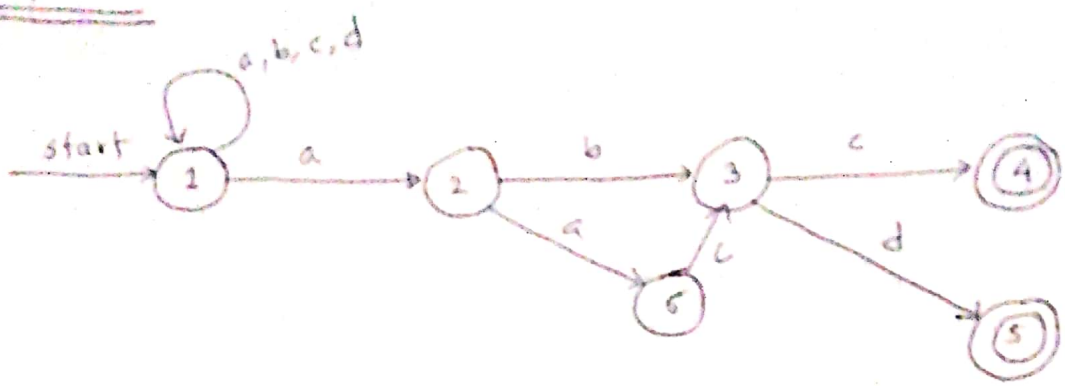
Ques:

1. Design NFA's to recognize the following sets of strings
   a) abc, abd, aacc and aacd. Assume the alphabet is {a, b, c, d}
   b) ab, bc, and ca. Assume the alphabet is {a, b, c}

2. Convert each of the NFAs of Question 1 into DFA's.

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1 a) Ans:



Conversion to DFA

Transition table for DFA:

	a	b	c	d
→ {1}	{1,2}	{1}	{1}	{1}
{1,2}	{1,2,6}	{1,3}	{1}	{1}
{1,2,6}	{1,2,6}	{1,3}	{1,3}	{1}
{1,3}	{1,2}	{1}	{1,4}	{1,5}
* {1,4}	{1,2}	{1}	{1}	{1}
* {1,5}	{1,2}	{1}	{1}	{1}

⇓

	a	b	c	d
→ A	B	A	A	A
B	C	D	A	A
C	C	D	D	A
D	B	A	E	F
* E	B	A	A	A
* F	B	A	A	A

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